

Beyond Targets: Spatial Spillovers of Targeted Poor County Program

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Abstract

This paper examines the spatial spillover effects of 2011 Targeted Poor County (TPC) program in China, which directs central government fiscal transfers to designated poor counties for investments in infrastructure, industrial development, and worker training. We extend a difference-in-differences framework to distinguish direct effects in target counties from spillover effects in adjacent non-target counties. Using household-level data from the National Fixed Point Survey and county-level statistical data, we find that household income increased and that households in both target and adjacent non-target counties reallocated labor from agricultural to nonfarm activities following the program. We provide evidence consistent with three channels underlying these spillover effects: improved transportation expanding cross-county access to nonfarm employment opportunities, industrial expansion generating local agglomeration spillovers, and greater access to agricultural machinery services reducing agricultural labor demand. These findings show that place-based development policies can generate economically meaningful effects beyond target administrative boundaries.

Keywords: place-based development; spatial spillovers; rural labor reallocation; poverty alleviation

JEL Codes: O12, O18, J21, R11

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1 Introduction

Development policies are typically evaluated within target populations or regions, although spillovers to untreated areas may substantially alter their aggregate impacts (Kline and Moretti, 2014; Egger et al., 2022). In 2011, China launched a new round of the targeted poor county (TPC) program, directing fiscal resources to designated poor counties. Huang and You (2025) document higher farm and nonfarm income and rural labor reallocation from farm to nonfarm sectors within target counties. Evidence from other place-based policies suggests spatial spillovers can be substantial, yet whether households in nearby non-target counties also reallocate labor from farm to nonfarm activities in response to spillovers from the TPC program remains largely unexplored.

In this study, we extend the standard difference-in-differences (DID) framework by distinguishing target counties from adjacent non-target counties, defined as non-target counties sharing an administrative border with at least one target county, to identify spatial spillovers of the 2011 TPC program. Using the National Fixed Point Survey (NFPS) and China County Statistical Yearbook, we find that households residing in adjacent non-target counties also reallocated labor from farm to nonfarm activities following the program, indicating spillovers beyond administrative boundaries.

We show that nonfarm activity spillovers work through three channels. First, improved transportation in target counties improves cross-county access to nonfarm employment opportunities for households in adjacent counties. Second, industrial expansion in target counties generates agglomeration spillovers that stimulate firm growth and employment in neighboring areas. Third, mechanization in target counties extends through cross-county machinery services, reducing agricultural labor demand and promoting nonfarm labor reallocation.

This paper directly extends Huang and You (2025) by documenting spatial spillovers of the TPC program to adjacent non-target counties. It also contributes to the place-based policy literature by showing that treatment effects can extend across administrative boundaries, underscoring the importance of accounting for spatial spillovers in policy evaluation (Ehrlich and Overman, 2020; Garin and Rothbaum, 2025).

2 Policy Background and Conceptualization

The TPC program has undergone several rounds of county designation since the mid-1980s. Following [Huang and You \(2025\)](#), we focus on the 2011 round of the national TPC program, under which 832 counties were designated primarily according to local economic conditions.¹

Target counties received central fiscal transfers for infrastructure, industrial development, and worker training. [Huang and You \(2025\)](#) document higher nonfarm income and activity in target counties, alongside improvements in local infrastructure, industrial development, and agricultural mechanization. We examine whether adjacent non-target counties experienced similar growth in nonfarm activities through three potential channels:

Transport and worker mobility. Improved transportation infrastructure in target counties facilitates cross-county mobility for workers in adjacent non-target counties seeking nonfarm employment opportunities ([Morten and Oliveira, 2024](#)).

Firm agglomeration. Industrial expansion directly induced by the TPC program in target counties may generate agglomeration spillovers that stimulate firm growth and nonfarm employment in neighboring areas ([Ellison et al., 2010](#); [Greenstone et al., 2010](#)).

Agricultural machinery services. The TPC program promotes agricultural mechanization in target counties, while improved transportation facilitates the cross-county provision of machinery services to neighboring counties, reducing agricultural labor demand and promoting labor reallocation toward nonfarm activities ([Ma et al., 2024](#)).

3 Data and Empirical Strategy

3.1 Data

We include two datasets on the county and household levels. County-level variables are drawn from the China County Statistical Yearbook from 2006 to 2019 to align with [Huang and You \(2025\)](#). Household-level data come from NFPS, which is the largest rural household-tracking survey in China. We use the NFPS sample from 2006 to 2015 due to the restriction on data access. For the mechanism analysis, we add variables related with transportation network and farmland terrain

¹Earlier rounds were initiated in 1986, 1994, and 2001; the 2001 list was subsequently subject to a dynamic adjustment round in 2006.

characteristics from [Davis et al. \(2025\)](#). The descriptive statistics are presented in Appendix Table [A1](#).

3.2 Empirical Strategy

We extend the standard DID specification by including non-target counties that share an administrative border with target counties to estimate spatial spillovers. For household-level outcomes, we estimate:

$$y_{it} = \beta^{target} target_{it} + \beta^{adj} adj_{it} + \delta_i + \lambda_{p,t} + \sum_{\tau} \gamma^{\tau} (pastRound_i^{\tau} \times t) + \varepsilon_{it}, \quad (1)$$

where $target_{it}$ equals one if household i resides in a target county with $t \geq 2011$; adj_{it} equals one if household i resides in an adjacent non-target county sharing an administrative border with at least one target county and $t \geq 2011$. δ_i and $\lambda_{p,t}$ denote household and province-year fixed effects. We further allow counties designated in 2001 or included in the 2006 adjustment round to follow round-specific linear trends by $pastRound_i^{\tau} \times t$. The coefficients β^{target} and β^{adj} capture the direct and spillover effects of the TPC program, respectively.

To estimate the dynamic effects and provide suggestive evidence on the parallel-trend assumption, we estimate the following event-study specification:

$$y_{it} = \sum_{k \neq -1} \beta_k^{target} D_{k,it}^{target} + \sum_{k \neq -1} \beta_k^{adj} D_{k,it}^{adj} + \delta_i + \lambda_{p,t} + \sum_{\tau} \gamma^{\tau} (pastRound_i^{\tau} \times t) + \varepsilon_{it} \quad (2)$$

where $D_{k,it}^{target}$ and $D_{k,it}^{adj}$ indicate the relative year k to year 2011 for households in target and adjacent non-target counties, respectively. The coefficients β_k^{target} and β_k^{adj} are relative to the omitted period $k = -1$; insignificant pre-treatment estimates support the parallel-trend assumption.

For county-level data, we use the same spillover-DID and event-study specifications but replace the household FE by county FE.

4 Results

4.1 Baseline Results

Table 1 reports the baseline estimates of the direct and spillover effects using the DID specification. Columns (1) to (3) show the program significantly increased GDP per capita, farm income, and nonfarm income in target counties by approximately 15.0%, 13.5%, and 18.8%, respectively, with corresponding increases of 10.7%, 9.6%, and 10.7% in adjacent non-target counties.

Columns (4) to (6) further decompose these economic changes through shifts in rural labor allocation. In target counties, farm labor input days declined by approximately 9.4%, nonfarm labor input days increased by 23.5%, and the county-level share of rural nonfarm workers rose by 12.2%. Similar changes occurred in adjacent non-target counties, where farm labor input declined by 5.4%, nonfarm labor input increased by 19.1%, and the rural nonfarm worker share rose by 11.1%. These results indicate substantial spillover effects on labor reallocation from farm to nonfarm activities in adjacent non-target counties.

Figure 1 reports the event-study estimates, which shows insignificant pre-treatment coefficients with post-treatment estimates broadly consistent with the baseline DID results.

4.2 Mechanism and Heterogeneity Analysis

In Tables 2 and 3, we examine three potential channels through which changes induced by the TPC program in target counties may have affected adjacent non-target counties.

Table 2 first examines the outcomes associated with the three proposed channels. Columns (1) and (2) show that highway and railway density increased by 23.2% and 11.9% in target counties, consistent with the TPC program’s investment in transportation infrastructure. Columns (3) and (4) show that industry-sector value added and the number of large industrial firms increased by 16.8% and 13.0% in target counties. Similar increases are observed in adjacent counties, at 8.9% and 5.0%, respectively, consistent with agglomeration spillovers in local industrial activity.

Columns (5) and (6) show that machinery ownership increased by 6.1% in target counties, while machinery use rose by 10.5% in target counties and 8.7% in adjacent counties. The increase in machinery use in adjacent counties was not accompanied by a corresponding rise in machinery ownership. This finding suggests that machinery acquired in target counties from TPC may have

been accessed through cross-county service provision (e.g., rental services).

Figure 2 of the event-study estimates shows no significant pre-trends and post-treatment estimates consistent with the DID results.

We next use Table 3 to provide more evidence on the three proposed channels by examining how the labor-reallocation effects vary with relevant baseline conditions. In Columns (1) to (3), the interactions with baseline highway density, railway density, and the number of industrial firms are significantly negative for both target and adjacent counties, indicating stronger shifts toward nonfarm labor where transport infrastructure and industrial development were initially weaker. Columns (4) and (5) show smaller increases in household nonfarm labor input in areas with greater farmland ruggedness and fragmentation,² which constrain machinery use (Afridi et al., 2023; Ma et al., 2024). These findings are further consistent with the three proposed mechanisms.

Appendix B discusses heterogeneous effects, showing stronger labor-reallocation effects among households with more working-age members and greater pre-existing family-business engagement, with similar estimates in target and adjacent counties.

4.3 Robustness Checks

Second-order spillovers. In Appendix Tables C1 and C2, we examine whether spillovers extend beyond directly adjacent counties by adding second-order neighboring counties, defined as non-target counties adjacent to first-order neighboring counties but not directly adjacent to target counties, and find no significant effects beyond directly adjacent counties.

Confounding policies and regional characteristics. In Appendix Tables C3 and C4, excluding counties affected by concurrent place-based policies (contiguous destitute areas, and old revolutionary base areas) or distinctive regional characteristics (Xinjiang and Tibet, and ethnic minority counties) leaves the estimates largely unchanged.

Selection into TPC. In Appendix Tables C5 and C6, allowing differential linear trends by pre-treatment economic, infrastructure, and agricultural conditions produces similar results.

Anticipation effect. In Appendix Figures C1 and C2, placebo treatment years prior to 2011 yield coefficients close to zero, providing additional evidence against pre-existing trends and

²Farmland ruggedness is measured at the county level as the standard deviation of within-county elevations, following Davis et al. (2025). Farmland fragmentation is measured at the household level from NFPS as the inverse of the average farmland area per plot, such that a higher value indicates more fragmented farmland.

anticipation effects.

5 Conclusion

This paper extends [Huang and You \(2025\)](#) by examining the spatial spillover effects of China’s 2011 TPC program on rural labor allocation. We find that the program increased household income and shifted labor from agricultural to nonfarm activities not only in target counties but also in adjacent non-target counties. The evidence suggests that improved transportation, industrial expansion, and greater access to agricultural machinery services contribute to these spillover effects. Overall, the findings show that the effects of place-based development policies can extend beyond directly target regions, highlighting the importance of accounting for spatial spillovers in policy evaluation.

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Tables

Table 1: Direct and Spillover Effects on Local Growth and Rural Labor Allocation

Dep. var.:	Growth & rural household income			Labor allocation		
	County GDP per capita log (1)	Farm income per member log (2)	Nonfarm income per member log (3)	Household farm labor input days log (4)	Household nonfarm labor input days log (5)	County rural nonfarm labor share (6)
$target_{it}$	0.140*** (0.026)	0.127** (0.059)	0.172*** (0.062)	-0.099** (0.046)	0.211*** (0.085)	0.115*** (0.010)
adj_{it}	0.102*** (0.018)	0.092** (0.041)	0.102** (0.048)	-0.055** (0.024)	0.175*** (0.058)	0.105*** (0.008)
Household FE	N	Y	Y	Y	Y	N
County FE	Y	N	N	N	N	Y
Province-year FE	Y	Y	Y	Y	Y	Y
Past policy trend	Y	Y	Y	Y	Y	Y
N	21,449	100,615	100,615	100,615	100,615	21,449
R-squared	0.903	0.778	0.695	0.757	0.670	0.826

Note: Column (1) and (6) use the county-level data. Columns (2) to (5) use the household-level data. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

Table 2: Effects on Infrastructure, Local Industrial Development, and Mechanization

Dep. var.:	Road infrastructure		Local economic development		Farm mechanization	
	Highway density log	Railway density log	Industry-sector value added per capita log	Large industrial firm number log	Machinery use per farmland log	Machinery owned per farmland log
	(1)	(2)	(3)	(4)	(5)	(6)
<i>target_{it}</i>	0.209*** (0.063)	0.112*** (0.043)	0.155*** (0.040)	0.122*** (0.036)	0.100*** (0.014)	0.059*** (0.019)
<i>adj_{it}</i>	0.025 (0.019)	0.038 (0.035)	0.085*** (0.024)	0.049** (0.023)	0.083*** (0.015)	0.034 (0.021)
Household FE	N	N	N	N	N	Y
County FE	Y	Y	Y	Y	Y	N
Province-year FE	Y	Y	Y	Y	Y	Y
Past policy trend	Y	Y	Y	Y	Y	Y
N	21,449	21,449	21,449	21,449	21,449	100,615
R-squared	0.857	0.954	0.920	0.898	0.878	0.703

Note: Columns (1) to (5) use county-level data. Column (6) uses household-level data. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

Table 3: Mechanism Analysis

Dep. var.:	County rural nonfarm labor share			Household nonfarm labor input days log	
Interaction var.:	Road infrastructure		Firm agglomeration	Farm mechanization	
	Standardized baseline highway density (1)	Standardized baseline railway density (2)	Standardized baseline industrial firm number (3)	Standardized baseline farmland ruggedness (4)	Standardized baseline farmland fragmentation (5)
$target_{it} \times interaction_c$	-0.028** (0.013)	-0.018** (0.009)	-0.016** (0.007)	-0.011** (0.005)	-0.007** (0.003)
$adj_{it} \times interaction_c$	-0.034** (0.014)	-0.023** (0.010)	-0.021** (0.009)	-0.018** (0.008)	-0.056** (0.026)
$target_{it}$	0.118*** (0.031)	0.116*** (0.031)	0.113*** (0.025)	0.484*** (0.183)	0.394*** (0.145)
adj_{it}	0.108*** (0.030)	0.106*** (0.030)	0.105*** (0.030)	0.414*** (0.124)	0.314*** (0.120)
Household FE	N	N	N	Y	Y
County FE	Y	Y	Y	N	N
Province-year FE	Y	Y	Y	Y	Y
Past policy trend	Y	Y	Y	Y	Y
N	21,449	21,449	21,449	100,615	100,615
R-squared	0.899	0.882	0.875	0.679	0.680

Note: Columns (1) to (3) use county-level data. Columns (4) and (5) use household-level data. All the interaction variables are measured by the pre-treatment average level and are standardized to improve comparability. For $adj_{it} \times interaction_c$ in Columns (1) and (2), highway and railway density refer to the corresponding measures in neighboring target counties. Farmland ruggedness is measured at the county level as the standard deviation of within-county elevations, following [Davis et al. \(2025\)](#). Farmland fragmentation is measured as the inverse of the average farmland area per plot for each household. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

Figures

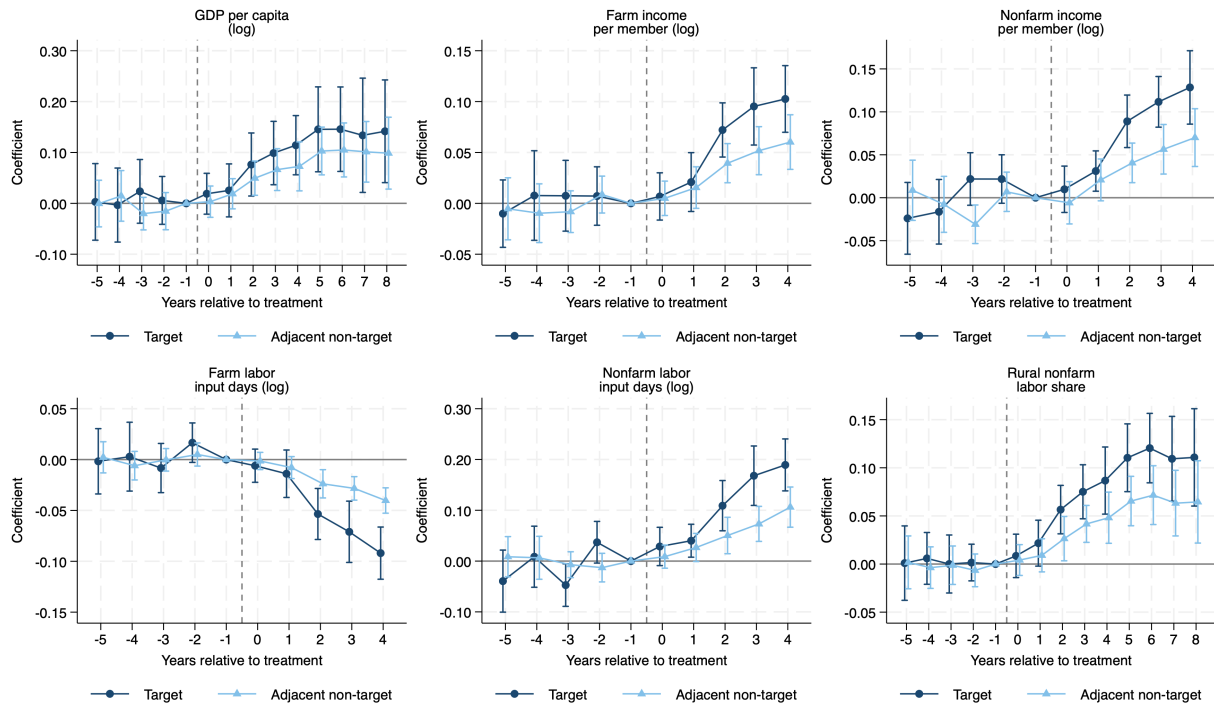


Figure 1: Event-Study Estimates of Main Effects

Note: This figure plots the estimated coefficients of event-study specification of main effects in Table 1.

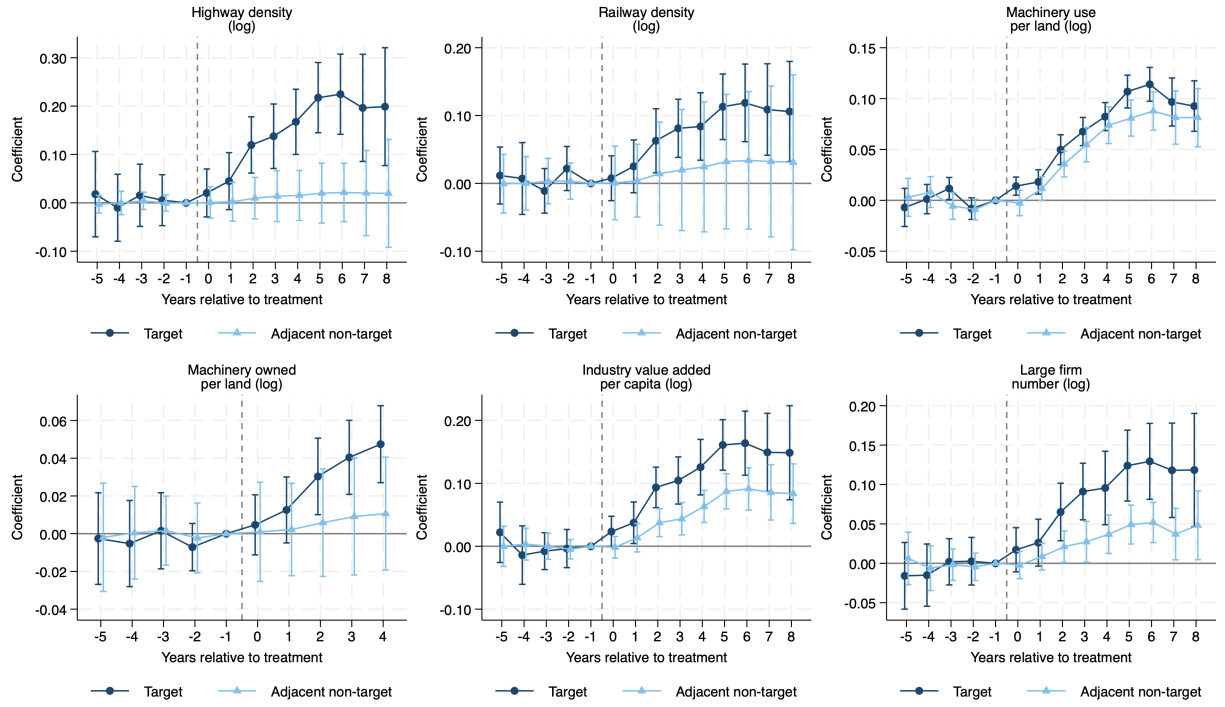


Figure 2: Event-Study Estimates of Mechanism Analysis

Note: This figure plots the estimated coefficients of event-study specification of mechanism in Table 2.

Appendix

A Descriptive Statistics

Table A1: Descriptive Statistics

Variables	Target	Adjacent non-target	Control
NFPS: household variables			
Household farm income per year (CNY)	4,136.32 (5,470.44)	4,435.89 (5,651.04)	3,850.99 (6,059.77)
Household nonfarm income per year (CNY)	7,967.83 (12,100.83)	8,271.90 (12,244.29)	9,673.14 (14,687.65)
Household farm labor input days per year	66.92 (88.04)	62.06 (101.80)	37.90 (90.20)
Household nonfarm labor input days per year	65.99 (82.24)	65.24 (81.00)	55.56 (78.41)
Number of farm machinery owned	0.27 (2.13)	0.31 (2.11)	0.24 (0.99)
Power of farm machinery owned (kW)	4.45 (148.77)	337.60 (7,227.73)	1.41 (10.01)
Number of family members	5.35 (2.04)	5.81 (2.80)	4.08 (2.77)
Number of family labor force	4.02 (1.52)	3.89 (1.44)	3.80 (1.45)
Area of farmland owned (mu)	6.46 (10.05)	5.52 (8.99)	4.59 (9.94)
Area per farmland plot (mu)	1.90 (1.11)	2.05 (1.18)	2.07 (1.27)
Observation	10,061	11,438	15,357
County statistical books: county variables			
GDP per capita (million CNY)	91.65 (78.86)	177.21 (153.32)	252.11 (210.01)
Population (thousand)	383.33 (299.42)	492.90 (312.12)	566.34 (339.50)
Agriculture-sector value added per capita	8.27 (7.52)	14.54 (10.70)	18.35 (12.85)
Industry-sector value added per capita	12.33 (15.25)	37.52 (41.98)	73.29 (81.93)
Service-sector value added per capita	10.27 (9.32)	22.66 (19.90)	45.46 (55.34)
Share of nonfarm rural workers	0.29 (0.15)	0.37 (0.15)	0.47 (0.18)
Power of farm machinery use (thousand kWh)	226.81 (250.56)	388.82 (342.27)	402.20 (326.09)
Number of large industrial firm	27.35 (28.32)	70.11 (71.30)	192.62 (211.91)
Highway density	11.23 (24.47)	33.15 (45.56)	59.38 (70.21)
Railway density	11.14 (19.50)	28.61 (55.78)	42.34 (88.04)
Terrain ruggedness	1,407.16 (965.39)	1,045.09 (1,012.95)	661.59 (670.08)
Group size	748	661	976

Note: This table reports the mean of each variables before 2011. Standard deviations are in parentheses. The number of target counties is less than 832 due to the missing information from the County Statistical Yearbook.

B Heterogeneity Analysis

To better understand the distributional consequences, Appendix Table B1 examines heterogeneity in the effects of the TPC program on household nonfarm labor input. Column (1) shows that the interaction with the family labor force share is significantly positive for both target and adjacent counties, indicating stronger increases in nonfarm labor input among households with a larger share of working-age members. Column (2) shows that the interaction with an indicator for family business as the primary income source is also significantly positive in both groups. This suggests that households more engaged in family business experienced larger increases in nonfarm labor input following the TPC program. Overall, the results indicate that both the direct and spillover effects on labor reallocation vary systematically with household labor availability and pre-existing engagement in nonfarm economic activities.

Table B1: Heterogeneity Analysis

Dep. var.:	Household nonfarm labor input days log	
Interaction var.:	Family labor force share (1)	Family business as primary income source (2)
$target_{it} \times interaction_c$	0.167** (0.069)	0.048*** (0.018)
$adj_{it} \times interaction_c$	0.101** (0.046)	0.018** (0.008)
$target_{it}$	0.278** (0.118)	0.311** (0.135)
adj_{it}	0.268*** (0.098)	0.294*** (0.096)
Household FE	Y	Y
Province-year FE	Y	Y
Past policy trend	Y	Y
N	100,615	100,615
R-squared	0.681	0.680

Note: This table uses household-level data. All the interaction variables are measured by the pre-treatment average level and are standardized to improve comparability. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

C Robustness

Table C1: Spillover Effects on Second-Order Counties: Main Effects

Dep. var.:	Local growth & rural household income			Labor allocation		
	County GDP per capita log (1)	Farm income per member log (2)	Nonfarm income per member log (3)	Household farm labor input days log (4)	Household nonfarm labor input days log (5)	County rural nonfarm labor share (6)
$target_{it}$	0.151*** (0.026)	0.152** (0.068)	0.181*** (0.061)	-0.102** (0.047)	0.221*** (0.081)	0.123*** (0.030)
adj_{it}	0.111*** (0.028)	0.098** (0.043)	0.113** (0.050)	-0.067** (0.029)	0.184*** (0.060)	0.112*** (0.013)
$second_{it}$	0.002 (0.002)	0.003 (0.006)	0.004 (0.005)	0.004 (0.006)	0.007 (0.012)	0.002 (0.004)
Household FE	N	Y	Y	Y	Y	N
County FE	Y	N	N	N	N	Y
Province-year FE	Y	Y	Y	Y	Y	Y
Past policy trend	Y	Y	Y	Y	Y	Y
N	21,449	100,615	100,615	100,615	100,615	21,449
R-squared	0.912	0.802	0.723	0.781	0.692	0.854

Note: The indicator $second_{it}$ equals to one for counties sharing boundaries with adj_{it} but not with $target_{it}$. Columns (1) and (6) use county-level data. Columns (2) to (5) use household-level data. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

Table C2: Spillover Effects on Second-Order Counties: Mechanism

Dep. var.:	Road infrastructure		Farm mechanization		Local economic development	
	Highway density log	Railway density log	Machinery per farmland use log	Machinery per farmland owned log	Industry-sector value added per capita log	Large industrial firm number log
	(1)	(2)	(3)	(4)	(5)	(6)
<i>target_{it}</i>	0.218*** (0.063)	0.122*** (0.043)	0.117*** (0.014)	0.092*** (0.019)	0.171*** (0.040)	0.145*** (0.036)
<i>adj_{it}</i>	0.035 (0.019)	0.048 (0.035)	0.098*** (0.015)	0.047 (0.021)	0.095*** (0.024)	0.067** (0.023)
<i>second_{it}</i>	0.000 (0.003)	0.001 (0.005)	0.002 (0.004)	0.000 (0.007)	0.000 (0.004)	0.001 (0.006)
Household FE	N	N	N	Y	N	N
County FE	Y	Y	Y	N	Y	Y
Province-year FE	Y	Y	Y	Y	Y	Y
Past policy trend	Y	Y	Y	Y	Y	Y
N	21,449	21,449	21,449	100,615	21,449	21,449
R-squared	0.894	0.961	0.887	0.763	0.948	0.924

Note: The indicator *second_{it}* equals to one for counties sharing boundaries with *adj_{it}* but not with *target_{it}*. Columns (1) to (3) and (5) to (6) use county-level data. Column (4) uses household-level data. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

Table C3: Exclude Counties Affected by Confounding Policies: Main Effects

Dep. var.:	Local growth & rural household income			Labor allocation		
	County GDP	Farm income	Nonfarm income	Household	Household	County rural
	per capita log (1)	per member log (2)	per member log (3)	farm labor input days log (4)	nonfarm labor input days log (5)	nonfarm labor share (6)
Panel A: exclude contiguous destitute areas						
<i>target_{it}</i>	0.153*** (0.028)	0.151** (0.067)	0.183*** (0.060)	-0.104** (0.046)	0.226*** (0.079)	0.124*** (0.032)
<i>adj_{it}</i>	0.109*** (0.027)	0.101** (0.044)	0.116** (0.051)	-0.069** (0.030)	0.188*** (0.061)	0.113*** (0.015)
FEs & trend	Y	Y	Y	Y	Y	Y
N	14,836	80,607	80,607	80,607	80,607	14,836
R-squared	0.875	0.615	0.647	0.714	0.657	0.804
Panel B: exclude Xinjiang and Tibet counties						
<i>target_{it}</i>	0.146*** (0.027)	0.159** (0.071)	0.174*** (0.058)	-0.109** (0.049)	0.233*** (0.085)	0.119*** (0.031)
<i>adj_{it}</i>	0.116*** (0.029)	0.092** (0.041)	0.121** (0.052)	-0.064** (0.028)	0.176*** (0.058)	0.108*** (0.014)
FEs & trend	Y	Y	Y	Y	Y	Y
N	20,211	98,202	98,202	98,202	98,202	20,211
R-squared	0.912	0.631	0.696	0.756	0.671	0.825
Panel C: exclude ethnic minority counties						
<i>target_{it}</i>	0.149*** (0.026)	0.163** (0.073)	0.169*** (0.056)	-0.112** (0.050)	0.239*** (0.083)	0.116*** (0.030)
<i>adj_{it}</i>	0.121*** (0.030)	0.095** (0.042)	0.126** (0.054)	-0.061** (0.027)	0.181*** (0.059)	0.105*** (0.013)
FEs & trend	Y	Y	Y	Y	Y	Y
N	19,942	96,248	96,248	96,248	96,248	19,942
R-squared	0.826	0.649	0.702	0.724	0.665	0.814
Panel D: exclude old revolutionary base areas						
<i>target_{it}</i>	0.145*** (0.026)	0.134** (0.062)	0.175*** (0.065)	-0.103** (0.047)	0.225** (0.095)	0.118*** (0.010)
<i>adj_{it}</i>	0.105*** (0.020)	0.100** (0.044)	0.110** (0.053)	-0.055** (0.024)	0.178*** (0.060)	0.110*** (0.008)
FEs & trend	Y	Y	Y	Y	Y	Y
N	17,409	97,254	97,254	97,254	97,254	17,409
R-squared	0.825	0.642	0.701	0.721	0.665	0.817

Note: Columns (1) and (6) use county-level data. Columns (2) to (5) use household-level data. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

Table C4: Exclude Counties Affected by Confounding Policies: Mechanism

Dep. var.:	Road infrastructure		Farm mechanization		Local economic development	
	Highway density log	Railway density log	Machinery per farmland use log	Machinery per farmland owned log	Industry-sector value added per capita log	Large industrial firm number log
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: exclude contiguous destitute areas						
<i>target_{it}</i>	0.197*** (0.060)	0.119*** (0.045)	0.096*** (0.013)	0.062*** (0.020)	0.148*** (0.039)	0.128*** (0.038)
<i>adj_{it}</i>	0.027 (0.021)	0.035 (0.034)	0.087*** (0.016)	0.032 (0.022)	0.081*** (0.023)	0.052** (0.024)
FEs & trend	Y	Y	Y	Y	Y	Y
N	14,836	14,836	14,836	80,607	14,836	14,836
R-squared	0.817	0.941	0.837	0.697	0.915	0.847
Panel B: exclude Xinjiang and Tibet counties						
<i>target_{it}</i>	0.215*** (0.064)	0.108*** (0.041)	0.104*** (0.015)	0.057*** (0.018)	0.161*** (0.041)	0.117*** (0.035)
<i>adj_{it}</i>	0.023 (0.020)	0.041 (0.036)	0.079*** (0.014)	0.036 (0.022)	0.088*** (0.025)	0.047** (0.022)
FEs & trend	Y	Y	Y	Y	Y	Y
N	20,211	20,211	20,211	98,202	20,211	20,211
R-squared	0.819	0.962	0.857	0.722	0.891	0.862
Panel C: exclude ethnic minority counties						
<i>target_{it}</i>	0.202*** (0.061)	0.116*** (0.044)	0.098*** (0.014)	0.061*** (0.019)	0.151*** (0.038)	0.125*** (0.037)
<i>adj_{it}</i>	0.028 (0.020)	0.036 (0.034)	0.086*** (0.016)	0.033 (0.021)	0.082*** (0.023)	0.051** (0.024)
FEs & trend	Y	Y	Y	Y	Y	Y
N	19,942	19,942	19,942	96,248	19,942	19,942
R-squared	0.842	0.938	0.865	0.671	0.909	0.881
Panel D: exclude old revolutionary base areas						
<i>target_{it}</i>	0.213*** (0.062)	0.109*** (0.042)	0.103*** (0.015)	0.056*** (0.018)	0.158*** (0.039)	0.119*** (0.035)
<i>adj_{it}</i>	0.024 (0.019)	0.040 (0.037)	0.080*** (0.015)	0.037 (0.022)	0.089*** (0.025)	0.046** (0.022)
FEs & trend	Y	Y	Y	Y	Y	Y
N	17,409	17,409	17,409	97,254	17,409	17,409
R-squared	0.831	0.969	0.845	0.681	0.934	0.873

Note: Columns (1) to (3) and (5) to (6) use county-level data. Column (4) uses household-level data

* : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

Table C5: Robustness to Additional Baseline-Selection Trends: Main Effects

Dep. var.:	Local growth & rural household income			Labor allocation		
	County GDP per capita log (1)	Farm income per member log (2)	Nonfarm income per member log (3)	Household farm labor input days log (4)	Household nonfarm labor input days log (5)	County rural nonfarm labor share (6)
Panel A: baseline economic development trends						
$target_{it}$	0.126*** (0.027)	0.113** (0.056)	0.154** (0.060)	-0.090** (0.044)	0.190** (0.081)	0.104*** (0.011)
adj_{it}	0.091*** (0.019)	0.081** (0.039)	0.090** (0.044)	-0.049** (0.023)	0.155*** (0.054)	0.094*** (0.009)
FEs & trend	Y	Y	Y	Y	Y	Y
N	21,449	100,615	100,615	100,615	100,615	21,449
R-squared	0.908	0.785	0.703	0.763	0.677	0.832
Panel B: baseline infrastructure trends						
$target_{it}$	0.124*** (0.027)	0.111** (0.055)	0.150** (0.059)	-0.088** (0.043)	0.186** (0.080)	0.102*** (0.011)
adj_{it}	0.089*** (0.018)	0.079** (0.038)	0.088** (0.043)	-0.048** (0.023)	0.151*** (0.053)	0.092*** (0.009)
FEs & trend	Y	Y	Y	Y	Y	Y
N	21,449	100,615	100,615	100,615	100,615	21,449
R-squared	0.907	0.784	0.702	0.762	0.676	0.831
Panel C: baseline agricultural production condition trends						
$target_{it}$	0.119*** (0.026)	0.106** (0.053)	0.144** (0.057)	-0.085** (0.042)	0.179** (0.078)	0.098*** (0.010)
adj_{it}	0.085*** (0.018)	0.075** (0.037)	0.084** (0.042)	-0.046** (0.022)	0.145*** (0.052)	0.088*** (0.009)
FEs & trend	Y	Y	Y	Y	Y	Y
N	21,449	100,615	100,615	100,615	100,615	21,449
R-squared	0.910	0.787	0.705	0.765	0.679	0.834
Panel D: full baseline covariate trends						
$target_{it}$	0.112*** (0.025)	0.099** (0.049)	0.135** (0.055)	-0.080** (0.040)	0.166** (0.074)	0.091*** (0.010)
adj_{it}	0.080*** (0.017)	0.070** (0.035)	0.078** (0.039)	-0.042** (0.021)	0.134*** (0.048)	0.082*** (0.008)
FEs & trend	Y	Y	Y	Y	Y	Y
N	21,449	100,615	100,615	100,615	100,615	21,449
R-squared	0.915	0.793	0.712	0.771	0.685	0.842

Note: Columns (1) and (6) use county-level data. Columns (2) to (5) use household-level data. Panel A controls for pre-2011 economic development trends, including baseline GDP, population, agriculture-sector value added, industry-sector value added, and service-sector value added interacted with a linear year trend. Panel B controls for baseline infrastructure trends, including highway density and railway density interacted with a linear year trend. Panel C controls for baseline agricultural production condition trends, including farm machinery use, terrain ruggedness, and farmland endowment interacted with a linear year trend. Panel D includes all baseline covariate trends. All specifications include the same fixed effects and past policy trends as the baseline specification. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

Table C6: Robustness to Additional Baseline-Selection Trends: Mechanism

Dep. var.:	Road infrastructure		Farm mechanization		Local economic development	
	Highway density log	Railway density log	Machinery per farmland use log	Machinery per farmland owned log	Industry-sector value added per capita log	Large industrial firm number log
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: baseline economic development trends						
<i>target_{it}</i>	0.188*** (0.060)	0.101** (0.040)	0.090*** (0.013)	0.053*** (0.018)	0.140*** (0.037)	0.110*** (0.034)
<i>adj_{it}</i>	0.021 (0.018)	0.033 (0.032)	0.074*** (0.014)	0.030 (0.019)	0.075*** (0.022)	0.043** (0.021)
FEs & trend	Y	Y	Y	Y	Y	Y
N	21,449	21,449	21,449	100,615	21,449	21,449
R-squared	0.864	0.958	0.885	0.710	0.927	0.904
Panel B: baseline infrastructure trends						
<i>target_{it}</i>	0.183*** (0.059)	0.098** (0.039)	0.089*** (0.013)	0.052*** (0.018)	0.137*** (0.036)	0.107*** (0.033)
<i>adj_{it}</i>	0.020 (0.018)	0.032 (0.031)	0.072*** (0.014)	0.029 (0.019)	0.073*** (0.021)	0.042** (0.021)
FEs & trend	Y	Y	Y	Y	Y	Y
N	21,449	21,449	21,449	100,615	21,449	21,449
R-squared	0.868	0.960	0.886	0.711	0.928	0.905
Panel C: baseline agricultural production condition trends						
<i>target_{it}</i>	0.178*** (0.057)	0.094** (0.038)	0.085*** (0.012)	0.050*** (0.017)	0.132*** (0.035)	0.103*** (0.032)
<i>adj_{it}</i>	0.019 (0.017)	0.030 (0.030)	0.069*** (0.013)	0.027 (0.018)	0.070*** (0.020)	0.040** (0.020)
FEs & trend	Y	Y	Y	Y	Y	Y
N	21,449	21,449	21,449	100,615	21,449	21,449
R-squared	0.868	0.960	0.889	0.714	0.930	0.907
Panel D: full baseline covariate trends						
<i>target_{it}</i>	0.166*** (0.054)	0.087** (0.036)	0.080*** (0.012)	0.047*** (0.016)	0.122*** (0.033)	0.094*** (0.030)
<i>adj_{it}</i>	0.017 (0.016)	0.027 (0.028)	0.062*** (0.012)	0.024 (0.017)	0.064*** (0.019)	0.035** (0.018)
FEs & trend	Y	Y	Y	Y	Y	Y
N	21,449	21,449	21,449	100,615	21,449	21,449
R-squared	0.876	0.965	0.895	0.720	0.936	0.913

Note: Columns (1) to (3) and (5) to (6) use county-level data. Column (4) uses household-level data. Panel A controls for pre-2011 economic development trends, including baseline GDP, population, agriculture-sector value added, industry-sector value added, and service-sector value added interacted with a linear year trend. Panel B controls for baseline infrastructure trends, including highway density and railway density interacted with a linear year trend. Panel C controls for baseline agricultural production condition trends, including farm machinery use, terrain ruggedness, and farmland endowment interacted with a linear year trend. Panel D includes all baseline covariate trends. All specifications include the same fixed effects and past policy trends as the baseline specification. * : $p < 0.1$; ** : $p < 0.05$; *** : $p < 0.01$. Standard errors in parentheses are clustered at the county level.

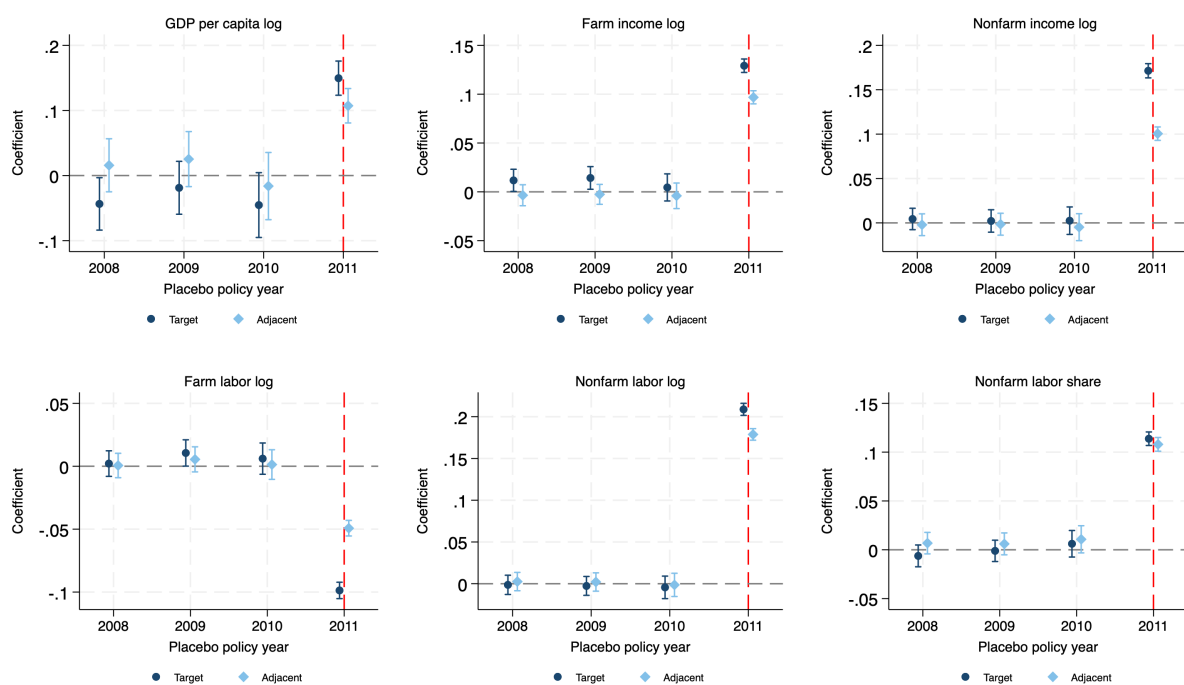


Figure C1: Anticipation: Main Effects

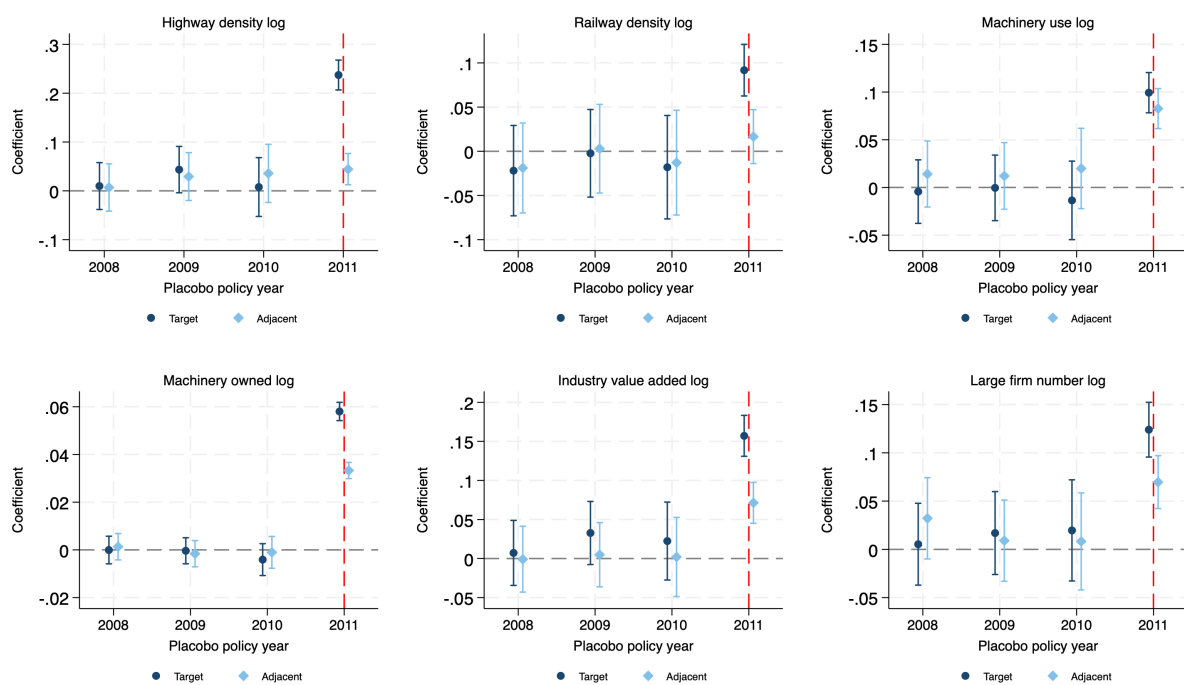


Figure C2: Anticipation: Mechanism Analysis